Detailed Syllabus
Lecture-wise Breakup

Course Code	**24B15CS314**	**Semester: ODD**	**Semester: 5th Session: 2026-27** Month from JUL to DEC
Course Name	Full Stack Development Lab		
Credits	1	**Contact Hours**	0-0-2 (L-T-P)

Faculty (Names)	**Coordinator(s)**	Dr. Sonal
	Teacher(s) (Alphabetically)	

COURSE OUTCOMES		**COGNITIVE LEVELS**
CO1	Explain Python programming concepts, Django architecture, and configure the development environment for web application development.	Understand (Level 2)
CO2	Build dynamic web applications using Django views, URL routing, templates, and template inheritance.	Apply (Level 3)
CO3	Implement database-driven web applications using Django ORM, models, CRUD operations, Forms, ModelForms, and model relationships.	Apply (Level 3)
CO4	Integrate user authentication, Django Admin, form validation, and data export features to enhance web application functionality.	Analyze (Level 4)
CO5	Evaluate and implement AJAX-based search and report generation features to enhance the functionality and usability of full-stack web applications.	Evaluate (Level 5)

Module No.	**Subtitle of the Module**	**Topics in the module**	**No. of Lectures for the module**
1.	Python and Django Fundamentals	Python programming for web development, Installation and configuration of Python, Visual Studio Code, Django, and virtual environments, Creating Django projects and applications, Understanding Django project structure and MVT architecture, Running the development server, Configuring project settings and installed applications.	2
2.	Views, URL Routing and Templates	Django request–response cycle, URL routing and URL configuration, creating function-based views, working with Django templates and Template Language (DTL), Passing context data from views to templates.	2
3.	Website Development and Template Inheritance	Developing multi-page websites, navigation between pages, template inheritance using base templates, Static file management (CSS, JavaScript, Images), creating reusable layouts, building responsive web pages.	2
4.	Database Applications using Django ORM	Introduction to Django ORM, creating models with various relationships, model relationships (one-to-one, one-to-many, many-to-many), CRUD operations, Forms and ModelForms. QuerySets, filtering and searching.	3
5.	Authentication and Administration	Introduction to Django Admin, register models in the admin interface, create and manage superuser and user accounts, implement user login and logout, create forms using Django Forms and ModelForms. apply basic form validation, restrict access to authenticated users.	2
6.	Reporting and Interactive Web Applications	Exporting model data to CSV, generating PDF reports, Integrating AJAX with Django, real-time search functionality, building interactive web applications.	3
Total number of Labs			**14**

Notional Hours

1. Assignment/Team Work (TW) /Self Learning (SL) (TW+SL) (in hours per semester) = 30 (PBL + Assignments+Evaluations+Quiz)

Total Notional Learning Hours = 30 hrs.

Evaluation Criteria

Components	**Maximum Marks**
Mid-Term	20
End Term Examination	20
D2D(Attendance/Assignment/Evaluation/PBL)	60
Total	**100**

Project based learning: Students shall develop a **mini full-stack web application** by integrating Python, Django, database operations, authentication, AJAX, and reporting features to solve a real-world problem in domains such as education, healthcare, inventory management, library management, or event management. The project should include planning, implementation, testing, documentation, and demonstration.

Recommended Reading material: Author(s), Title, Publisher, Edition, Year of Publication etc. (Text books, Reference Books, Journals, Reports, Websites etc. in the IEEE format)

Text Books

[1]	E. Matthes, *Python Crash Course*, 3rd ed. San Francisco, CA, USA: No Starch Press, 2023.
[2]	W. S. Vincent, *Django for Beginners: Build Websites with Python and Django*, 5th ed. Morrisville, NC, USA: WelcomeToCode, 2025.
[3]	A. Melé, *Django 5 By Example*, 5th ed. Birmingham, U.K.: Packt Publishing, 2024.

Reference Books:

[1]	M. C. Brown, *Python: The Complete Reference*, 4th ed. New York, NY, USA: McGraw-Hill Education, 2018.
[2]	D. Rubio, *Beginning Django: Web Application Development and Deployment with Python*, 2nd ed. Berkeley, CA, USA: Apress, 2017.
[3]	A. Holovaty and J. Kaplan-Moss, *The Definitive Guide to Django: Web Development Done Right*. Berkeley, CA, USA: Apress, 2009.

eLearning Resources:

1.	https://docs.djangoproject.com/en/6.0/
2.	https://docs.python.org/3/

CO-PO-PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**

CO1	3 (Applies Python programming and Django fundamentals to develop basic web application.)	1 (Identifies basic programming requirements and analyzes simple web application problems.)	1 (Develops simple web application components using Django.)		3 (Uses modern development tools such as Python, Django, VS Code, and virtual environments.)			1 (Performs laboratory exercises individually or collaboratively during application development.)			2 (Develops the ability to learn and adopt modern web development technologies independently.)	2 (Applies Python programming fundamentals and Django concepts to develop basic web applications, providing a foundation for designing software solutions.)	2 (Develops programming proficiency using Python and Django, preparing students for coding and programming challenges.)
CO2	2 (Applies web development concepts to develop dynamic web pages.)	2 (Analyzes user requirements to design multi-page web applications.)	3 (Designs and develops dynamic web applications using Django views, templates, and URL routing.)		3 (Effectively utilizes Django development tools and framework features.)	1 (Develops web applications that can support education, information sharing, and other societal services.)		1 (Collaborates in developing and integrating multiple web pages and templates.)		1 (Plans and organizes web application modules and development tasks.)	2 (Adapts to modern web development practices and frameworks.)	3 (Designs and develops dynamic web applications using appropriate web technologies and software development practices to address real-world problems.)	2 (Strengthens practical programming and web development skills useful for programming contests and technical competitions.)
CO3	2 (Applies database concepts to develop databa	2 (Analyzes application requirements to design databa	3 (Develops database-driven applications using Django	1 (Investigates application requirements throu	3 (Uses Django ORM and database management tools	1 (Develops applications for societal domains	1 (Applies ethical practices in handling user	1 (Collaborates in database design and application		1 (Organizes application modules and database compo	1 (Organizes application modules and database compo	3 (Develops database-driven web applications by selecting	3 (Enhances application development and programming

	se-driven web applications.)	se models and relationships.)	ORM, Forms, and Model Forms.)	gh database design and testing.)	effectively.)	such as education and healthcare.)	data and database operations.)	development.)		nents systematically.)	nents systematically.)	appropriate data models, ORM techniques, and software design approaches to solve real-world problems.)	skills required for project development and technical competitions.)
CO4	2 (Applies authentication and validation techniques in web applications.)	1 (Identifies security requirements for web applications.)	3 (Develops secure web applications using authentication, administration, and validation mechanisms.)		3 (Uses Django Admin, authentication, and validation tools effectively.)	1 (Develops secure applications that support reliable digital services for society.)	1 (Promotes ethical handling of user credentials and privacy.)	1 (Works collaboratively while integrating security and administration features.)		1 (Integrates application modules following systematic development practices.)	1 (Learns emerging techniques for secure web application development.)	2 (Implements authentication, administration, and validation mechanisms to improve the reliability and security of software solutions.)	3 (Strengthens practical software development skills through implementation of secure and scalable web applications.)
CO5	2 (Applies full-stack development concepts to	2 (Analyzes user requirements to imple	3 (Designs and develops complete interact	2 (Investigates and tests application functi	3 (Utilizes AJAX, Django, and reporting	1 (Develops interactive web applicatio	1 (Follows ethical software devel	2 (Functions effectively as a team member		1 (Plans, integrates, and manages application	3 (Demonstrates the ability to independently	3 (Designs and devel	3 (Develops advan

	build interactive web applications.)	ment AJAX-based interactive features.)	ive full-stack web applications.)	onality for dynamic data retrieval and reporting.)	tools for application development.)	ns for societal and community-oriented services.)	opment practices while handling user information.)	during integration, testing, and deployment.)		modules to complete the project successfully.)	learn and integrate emerging web technologies.)	ops complete interactive full-stack web applications integrating AJAX, reporting, and database technologies to address real-life applications.)	ced full-stack development skills that enable participation in programming, hackathons, and technology-driven project competitions.)

WK Mapping

CO	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9	CO-wise Justification
CO1										Students explain Python programming concepts, configure the Django environment, and apply core computing knowledge using modern development tools while strengthening problem-solving skills.
CO2										Students build dynamic web applications using Django views, URL routing, and templates by applying software engineering principles, modern development tools, and project-based learning for real-world applications.
CO3										Students implement database-driven web applications using Django ORM and CRUD operations while applying computing fundamentals, modern tools, testing practices, and ethical handling of application data.
CO4										Students integrate authentication, form validation, Django Admin, and data export features by applying advanced

										web development concepts, modern tools, testing methodologies, and secure software development practices.
CO5										Students evaluate and implement AJAX-based search and reporting features by integrating advanced web technologies, modern development tools, project-based learning, solution refinement, and ethical software development practices.